

3-Groups and higher gauge theory

unification of all interactions

Tijana Radenković
rtijana@ipb.ac.rs

Group for Gravitation, Particles and Fields,
Institute of Physics, University of Belgrade,
Pregrevica 118, 11080 Belgrade, Serbia

PhD Student Symposium 2021

1 Introduction

- Spinfoam quantization procedure
- 3-groups
- The $3BF$ theory

2 $3BF$ theory for the gravity and matter fields

- The group L
- The constrained $3BF$ action

3 Conclusions

- Within the LQG framework, the covariant quantization program is based on a "path integral on a lattice" idea, starting from the classical action for GR in the form of a BF theory with a simplicity constraint, known as the Plebanski action. The lattice quantization gives rise to spinfoam models.
- Since the Plebanski action does not contain the tetrad fields in its topological BF sector, the spinfoam models are hard to couple to matter fields. This problem is solved by passing to the $2BF$ action, which is a categorical generalization of the BF theory. The matter fields can now be coupled, and the lattice quantization is named "*spincube model*".
- The spincube model couples matter fields in the similar manner as it is done in Standard model. Another categorical generalization to the $3BF$ action combines gravity, gauge fields and matter fields in a unified geometric and algebraic way, paving the way to a "*categorical unification*" of all fields.

- The *covariant quantization* approach focuses on defining the gravitational path-integral of the theory:

$$Z = \int \mathcal{D}\phi e^{iS[\phi]}.$$

- This is done in three steps:
 - One write the action as the sum of the topological part and the simplicity constraints:

$$S[g] = S_{top}[g] + S_{con}[g].$$

- One defines the topological state sum using TQFT formalism:

$$Z_{top} = \int \mathcal{D}g e^{iS_{top}},$$

to be equal to the sum over states:

$$Z = \sum_{\{\phi\}} \prod_{v \in T} \mathcal{A}_v(\phi) \prod_{\epsilon \in T} \mathcal{A}_\epsilon(\phi) \cdots \prod_{\sigma \in T} \mathcal{A}_\sigma(\phi).$$

- Modifying the amplitudes of the topological state sum one obtains the state sum of the full theory:

$$Z = \int \mathcal{D}g e^{iS_{top} + iS_{con}}.$$

- A 3-group is equivalent to a **2-crossed module** $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, -\}_{\text{pf}})$:
 - Lie groups G, H , and L ,
 - the homomorphisms $\delta : L \rightarrow H$ and $\partial : H \rightarrow G$,
 - an action $\triangleright$ of the group G on all three groups,
 - and a G -equivariant map, called the Peiffer lifting:

$$\{-, -\}_{\text{pf}} : H \times H \rightarrow L,$$

and certain axioms are assumed to hold true among all these maps.

The main purpose of all this structure is to generalize the notion of parallel transport from curves to surfaces to volumes.

- Given a 2-crossed module, one can define a **3-connection**, an ordered triple (α, β, γ) , where α, β , and γ are algebra-valued differential forms,

$$\begin{aligned}\alpha &= \alpha^\alpha{}_\mu \tau_\alpha dx^\mu, & \alpha &\in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}), \\ \beta &= \beta^a{}_{\mu\nu} t_a dx^\mu \wedge dx^\nu, & \beta &\in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}), \\ \gamma &= \gamma^A{}_{\mu\nu\rho} T_A dx^\mu \wedge dx^\nu \wedge dx^\rho, & \gamma &\in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}).\end{aligned}\tag{1}$$

- Then introduce the line, surface and volume holonomies,

$$g = \mathcal{P}\exp \int_\gamma \alpha, \quad h = \mathcal{P}\exp \int_S \beta, \quad l = \mathcal{P}\exp \int_V \gamma.\tag{2}$$

- The corresponding **fake 3-curvature** $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined as:

$$\begin{aligned}\mathcal{F} &= d\alpha + \alpha \wedge \alpha - \partial\beta, & \mathcal{G} &= d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \\ \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \{\beta \wedge \beta\}_{\text{pf}}.\end{aligned}\tag{3}$$

At this point one can construct the so-called **3BF theory**.

- For a manifold $\mathcal{M}_4$ and the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, -\}_{\text{pf}})$, that gives rise to 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, one defines the 3BF action as

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}. \quad (4)$$

- 3BF theory is a topological gauge theory,
- it is based on the 3-group structure,
- it is a generalization of an ordinary BF theory for a given Lie group G .
- The physical interpretation of the Lagrange multipliers C and D :
 - **the $\mathfrak{h}$ -valued 1-form C can be interpreted as the tetrad field if $H = \mathbb{R}^4$ is the spacetime translation group,**

$$C \rightarrow e = e^a{}_{\mu}(x) t_a dx^{\mu}, \quad (5)$$

- **the $\mathfrak{l}$ -valued 0-form D can be interpreted as the set of real-valued matter fields, given some Lie group L :**

$$D \rightarrow \phi = \phi^A(x) T_A. \quad (6)$$

- 1 Introduction
⋮
- 2 **3BF theory for the gravity and matter fields**
⋮
- 3 Conclusions

- the (trivial) Standard Model 3-group:

-

$$G = SO(3, 1) \times SU(3) \times SU(2) \times U(1), \quad H = \mathbb{R}^4, \quad L - \text{to be discussed};$$

- boundary maps are trivial – for all $I \in L$ and $\vec{v} \in H$ we define

$$\delta I = 1H = 0, \quad \partial \vec{v} = 1G;$$

- the Peiffer lifting is trivial - for all $\vec{u}, \vec{v} \in H$ we define

$$\{\vec{u}, \vec{v}\}_{\text{pf}} = 1_L;$$

- the action of the group G on itself is via the adjoint representation;
- the action of the group G on H is via vector representation for the $SO(3, 1)$ sector and via trivial representation for the $SU(3) \times SU(2) \times U(1)$ sector;
- the action of the group G on L is nontrivial and depends on the choice of L (to be discussed).

One can verify that all axioms of 3-group are satisfied.

How do we choose the group L ?

- Given that $\phi = \phi^A T_A$, we have one real-valued field $\phi^A(x)$ for each generator $T_A \in L$.
- How many real-valued field components do we have in the matter sector of the Standard Model?**

1. lepton generation	red color 1. quark generation	green color 1. quark generation	blue color 1. quark generation
$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L$	$\begin{pmatrix} u_r \\ d_r \end{pmatrix}_L$	$\begin{pmatrix} u_g \\ d_g \end{pmatrix}_L$	$\begin{pmatrix} u_b \\ d_b \end{pmatrix}_L$
$(\nu_e)_R$	$(u_r)_R$	$(u_g)_R$	$(u_b)_R$
$(e^-)_R$	$(d_r)_R$	$(d_g)_R$	$(d_b)_R$

- **How many real-valued field components do we have in the matter sector of the Standard Model?**

- The fermion sector gives us:

$$16 \frac{\text{spinors}}{\text{family}} \times 3 \text{ families} \times 4 \frac{\text{real-valued fields}}{\text{spinor}} = 192 \text{ real-valued fields } \phi^A.$$

- The Higgs sector gives us:

$$2 \text{ complex scalar fields} = 4 \text{ real-valued fields } \phi^A.$$

- This suggests the structure for L in the form:

$$L = L_{\text{fermion}} \times L_{\text{Higgs}}, \quad \dim L_{\text{fermion}} = 192, \quad \dim L_{\text{Higgs}} = 4.$$

The action $G \triangleright L \rightarrow L$ specifies the transformation properties of each real-valued field ϕ^A with respect to Lorentz and internal symmetries.

- Moreover, G acts in the same way across families, suggesting the structure

$$L_{\text{fermion}} = L_{\text{1st family}} \times L_{\text{2nd family}} \times L_{\text{3rd family}}, \quad \dim L_{k\text{-th family}} = 64.$$

- We have formulated **the constrained 2BF actions** describing the Yang-Mills field and Einstein-Cartan gravity, and also **the constrained 3BF actions** describing the Klein-Gordon, Dirac, Weyl and Majorana fields coupled to gravity in the standard way.
- A realisation that reproduces the Standard Model:
 - The groups are:

$$G = SO(3,1) \times SU(3) \times SU(2) \times U(1), \quad H = \mathbb{R}^4,$$

$$L = \mathbb{R}^4(\mathbb{C}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}).$$

- The constrained 3BF action is:

$$\begin{aligned}
 S = & \int B_{\hat{\alpha}} \wedge \mathcal{F}^{\hat{\alpha}} + e_{\hat{a}} \wedge \mathcal{G}^{\hat{a}} + D_{\hat{A}} \wedge \mathcal{H}^{\hat{A}} \\
 & + \left(B_{\hat{\alpha}} - C_{\hat{\alpha}}^{\hat{\beta}} M_{cd\hat{\beta}} e^c \wedge e^d \right) \wedge \lambda^{\hat{\alpha}} - \left(\gamma_{\hat{A}} - e^a \wedge e^b \wedge e^c C_{\hat{A}}^{\hat{B}} M_{abc\hat{B}} \right) \wedge \lambda^{\hat{A}} \\
 & + \zeta^{ab}{}_{\hat{\alpha}} \wedge \left(M_{ab}{}^{\hat{\alpha}} \varepsilon^{cdef} e_c \wedge e_d \wedge e_e \wedge e_f - F^{\hat{\alpha}} \wedge e_c \wedge e_d \right) \\
 & + \zeta^{ab}{}_{\hat{A}} \wedge \left(M_{abc}{}^{\hat{A}} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - F^{\hat{A}} \wedge e_a \wedge e_b \right) \\
 & - \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \left(Y_{\hat{A}\hat{B}\hat{C}} D^{\hat{A}} D^{\hat{B}} D^{\hat{C}} + M_{\hat{A}\hat{B}} D^{\hat{A}} D^{\hat{B}} + L_{\hat{A}\hat{B}\hat{C}\hat{D}} D^{\hat{A}} D^{\hat{B}} D^{\hat{C}} D^{\hat{D}} \right) \\
 & - 4\pi i l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c D_{\hat{A}} T^{d\hat{A}}{}_{\hat{B}} D^{\hat{B}},
 \end{aligned} \tag{7}$$

- 1 Introduction
⋮
- 2 3BF theory for the gravity and matter fields
⋮
- 3 Conclusions

- Higher gauge theory represents a formalism where gravity, gauge fields, fermions and Higgs are treated on an equal footing.
- The underlying algebraic structure of a 3-group classifies all fundamental fields by specifying groups L, H, G and their maps $\partial, \delta, \triangleright$, and $\{-, -\}_{\text{pf}}$.
- This structure has natural **geometrical interpretation** of parallel transport along a curve, a surface, and a volume.
- The gauge group L **specifies the complete matter sector of the Standard Model** if one chooses

$$L = \mathbb{R}^4(\mathbb{C}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}).$$

- The action $\triangleright$ of G on L specifies the **transformation properties** of matter fields.
- Nontrivial choices of the 3-group structure may provide new avenues for research on unification of all fields.

- [1] T. Radenković and M. Vojinović, *Gauge symmetry of the 3BF theory for a generic Lie 3-group*, arXiv:2101.04049.
- [2] T. Radenković and M. Vojinović, *Higher Gauge Theories Based on 3-groups*, *J. High Energy Phys.* **10**, 222 (2019), arXiv:1904.07566.
- [3] T. Radenković and M. Vojinović, *Hamiltonian Analysis for the Scalar Electrodynamics as 3BF Theory*, *Symmetry* **12**, 620 (2020), arXiv:2004.06901.

Thank you for your attention!